# Binary Classification Model Performance Comparison: Evaluating the Accuracy of TabTransformers and Graph Neural Networks in Financial Fraud Detection

Conler Simmons, Steven Fernandes, PhD
Department of Computer Design & Journalism

## Background & Motivation

### Research Question

*How do attention-based and graph-based architectures perform in detecting financial fraud?*

- TabTransformer asks whether fraud lives in how features combine; GNN asks whether fraud spreads through how transactions connect

### Dataset

- Source: IEEE-CIS Fraud Detection dataset (Kaggle / Vesta Corporation) — two raw files merged on TransactionID
- 590,540 financial transactions over ~6 months
- 3.5% fraud rate — severe class imbalance
- 434 raw features → reduced to 70 (50 numeric + 20 categorical) via LightGBM importance ranking
- Same 70 features used by both models — ensures a fair, direct comparison
- Primary metric: PR-AUC — at 3.5% fraud, a model predicting "not fraud" on every transaction achieves 96.5% accuracy.
- PR-AUC measures the precision-recall tradeoff across all thresholds and is robust to class imbalance. Random baseline ≈ 0.035

### Experimental Setup

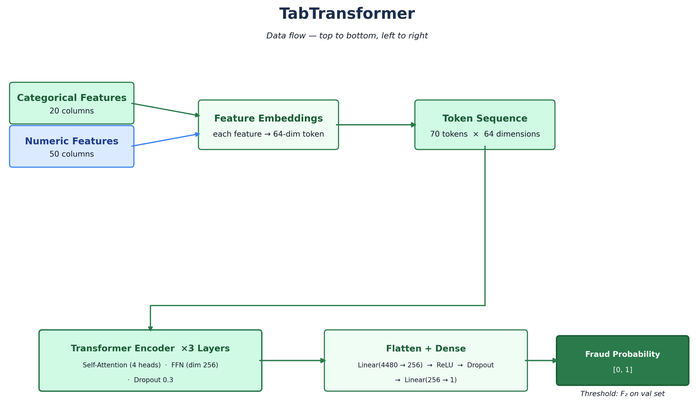
- Train / Validation / Test split: 70% / 15% / 15%, sorted by transaction time — models always train on older data and evaluate on newer, unseen transactions
- Decision threshold tuned on the validation set using F2 score, which weights recall twice as heavily as precision — in fraud detection, missing a fraud costs more than a false alarm
- Loss function: binary cross-entropy with ~27× class weighting to counteract the 3.5% fraud rate
- Label smoothing of 0.05 applied to both models to prevent overconfidence on noisy labels

### Model Perspectives

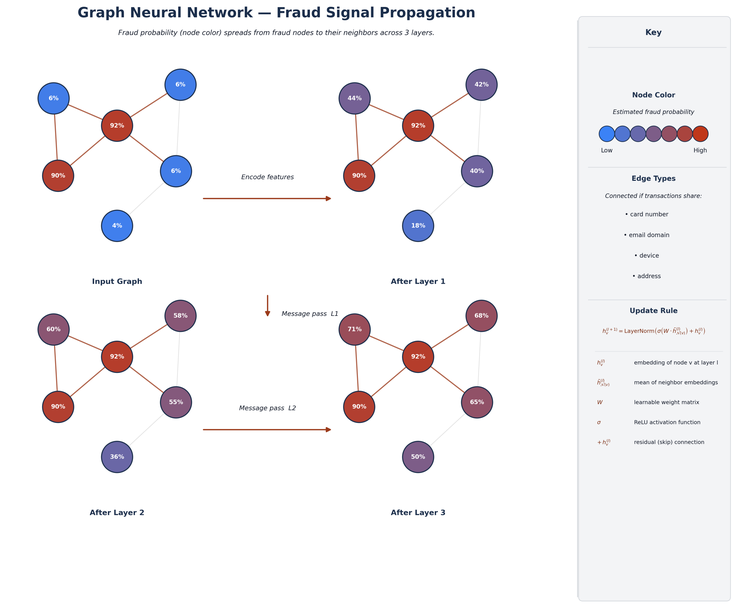
TabTransformer hypothesizes fraud lives in how features combine — attention captures joint patterns across cards, emails, and devices simultaneously. GNN hypothesizes fraud spreads through relationships — if a card was used fraudulently once, connected transactions deserve elevated suspicion. This dataset has both properties, making it an ideal testbed for a direct comparison.

*Both models implemented entirely from scratch — no pretrained weights, no external model libraries.*

## Model Architectures

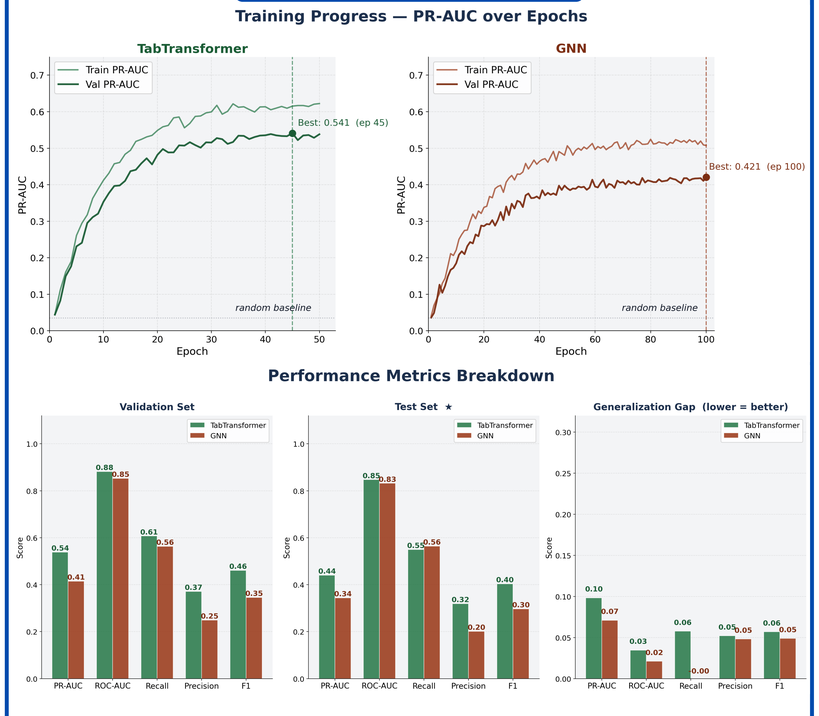



*Encodes every feature — categorical and numeric — as its own 64-dim token, then applies 3 layers of self-attention (4 heads) to discover which feature combinations signal fraud without being told which matter. Output: single fraud probability per transaction.*




*Constructs a transaction graph — edges link transactions sharing a card, email, device, or address — then propagates fraud signals outward across three rounds of message passing. Residual connections ensure each node retains its own identity through aggregation. Output: fraud probability per node.*

## Performance & Conclusion




## Results

### Head-to-Head Comparison

*TabTransformer vs. GNN · IEEE-CIS Fraud Detection · Test Set*

| Metric | TabTransformer | GNN | Winner |
|---|---|---|---|
| test PR-AUC ★ | **0.4402** | 0.3437 | TabTransformer |
| test ROC-AUC | **0.8471** | 0.8317 | TabTransformer |
| test Recall | 0.5495 | **0.5644** | GNN |
| test Precision | **0.3190** | 0.2012 | TabTransformer |
| test F1 | **0.4036** | 0.2967 | TabTransformer |
| Val→Test Gap ↓ | 0.098 | **0.071** | GNN |

★ *Primary metric* · ↓ *lower is better* · *Highlighted cell = winning value*

- Feature-level attention (TabTransformer) outperforms relational graph reasoning (GNN) on the primary metric — PR-AUC 0.44 vs. 0.34
- The GNN catches marginally more fraud (56% vs. 55% recall) but at the cost of significantly more false positives
- Both models perform 10–12× above the random baseline (PR-AUC ≈ 0.035) — meaningful signal exists in both approaches
- The two architectures are complementary — an ensemble combining feature-level attention with graph-level propagation is a natural direction for future work

*Vaswani et al. (2017). "Attention Is All You Need." NeurIPS. · Dataset: IEEE-CIS Fraud Detection, Kaggle / Vesta Corporation.*